

FORMULA TRIGO

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1.MESUREMENT OF ANGLE

$$(a) \frac{D}{180} = \frac{G}{200} = \frac{C}{\pi} \quad (b) \theta = \frac{l}{r}$$

2.TRIGONOMETRICAL FUNCTIONS

$$1.(a) \sin \theta = \frac{p}{h} \quad (b) \cos \theta = \frac{b}{h} \quad (c) \tan \theta = \frac{p}{b} \quad (d) \cot \theta = \frac{b}{p}$$

$$(e) \sec \theta = \frac{h}{p} \quad (f) \operatorname{cosec} \theta = \frac{h}{p}, \text{ Where } h^2 = p^2 + b^2$$

$$2.(a) \sin^2 \theta + \cos^2 \theta = 1 \quad (b) \sec^2 \theta - \tan^2 \theta = 1 \quad (c) \operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$(d) \tan \theta = \frac{\sin \theta}{\cos \theta} \quad (e) \cot \theta = \frac{\cos \theta}{\sin \theta} \quad (f) \sin \theta \cdot \operatorname{cosec} \theta = 1 \quad (g) \sec \theta \cdot \cos \theta = 1 \quad (h) \tan \theta \cdot \cot \theta = 1$$

3.(a) all trigonometrical functions are +ve in 1st quadrant.

(b) $\sin \theta$ and $\operatorname{cosec} \theta$ are +ve in 2nd quadrant.

(c) $\tan \theta$ and $\cot \theta$ are +ve in 3rd quadrant.

(d) $\cos \theta$ and $\sec \theta$ are +ve in 4th quadrant.

(e) T-functions of angle “-θ” in terms of “θ”

$$(i) \sin(-\theta) = -\sin \theta$$

$$(iii) \tan(-\theta) = -\tan \theta$$

$$(v) \sec(-\theta) = \sec \theta$$

$$(ii) \cos(-\theta) = \cos \theta$$

$$(iv) \cot(-\theta) = -\cot \theta$$

$$(vi) \operatorname{cosec}(-\theta) = -\operatorname{cosec} \theta$$

(f) T-functions of angle “90° - θ” in terms of “θ”

$$(i) \sin(90^\circ - \theta) = \cos \theta$$

$$(iii) \tan(90^\circ - \theta) = \cot \theta$$

$$(v) \sec(90^\circ - \theta) = \operatorname{cosec} \theta$$

$$(ii) \cos(90^\circ - \theta) = \sin \theta$$

$$(iv) \cot(90^\circ - \theta) = \tan \theta$$

$$(vi) \operatorname{cosec}(90^\circ - \theta) = \sec \theta$$

(g) T-functions of angle “90° + θ” in terms of “θ”

$$(i) \sin(90^\circ + \theta) = \cos \theta$$

$$(iii) \tan(90^\circ + \theta) = -\cot \theta$$

$$(v) \sec(90^\circ + \theta) = -\operatorname{cosec} \theta$$

$$(ii) \cos(90^\circ + \theta) = -\sin \theta$$

$$(iv) \cot(90^\circ + \theta) = -\tan \theta$$

$$(vi) \operatorname{cosec}(90^\circ + \theta) = \sec \theta$$

(h) T-functions of angle “180° - θ” in terms of “θ”

$$(i) \sin(180^\circ - \theta) = \sin \theta$$

$$(ii) \cos(180^\circ - \theta) = -\cos \theta$$

$$(iii) \tan(180^\circ - \theta) = -\tan\theta$$

$$(v) \sec(180^\circ - \theta) = -\sec\theta$$

$$(iv) \cot(180^\circ - \theta) = -\cot\theta$$

$$(vi) \operatorname{cosec}(180^\circ - \theta) = \operatorname{cosec}\theta$$

(i) T-functions of angle “ $180^\circ + \theta$ ” in terms of “ θ ”

$$(i) \sin(180^\circ + \theta) = -\sin\theta$$

$$(iii) \tan(180^\circ + \theta) = \tan\theta$$

$$(v) \sec(180^\circ + \theta) = -\sec\theta$$

$$(ii) \cos(180^\circ + \theta) = -\cos\theta$$

$$(iv) \cot(180^\circ + \theta) = \cot\theta$$

$$(vi) \operatorname{cosec}(180^\circ + \theta) = -\operatorname{cosec}\theta$$

(j) T-functions of angle “ $270^\circ - \theta$ ” in terms of “ θ ”

$$(i) \sin(270^\circ - \theta) = -\cos\theta$$

$$(iii) \tan(270^\circ - \theta) = \cot\theta$$

$$(v) \sec(270^\circ - \theta) = -\operatorname{cosec}\theta$$

$$(ii) \cos(270^\circ - \theta) = -\sin\theta$$

$$(iv) \cot(270^\circ - \theta) = \tan\theta$$

$$(vi) \operatorname{cosec}(270^\circ - \theta) = -\sec\theta$$

(k) T-functions of angle “ $270^\circ + \theta$ ” in terms of “ θ ”

$$(i) \sin(270^\circ + \theta) = -\cos\theta$$

$$(iii) \tan(270^\circ + \theta) = -\cot\theta$$

$$(v) \sec(270^\circ + \theta) = \operatorname{cosec}\theta$$

$$(ii) \cos(270^\circ + \theta) = \sin\theta$$

$$(iv) \cot(270^\circ + \theta) = -\tan\theta$$

$$(vi) \operatorname{cosec}(270^\circ + \theta) = -\sec\theta$$

(l) T-functions of angle “ $360^\circ - \theta$ ” in terms of “ θ ”

$$(i) \sin(360^\circ - \theta) = -\sin\theta$$

$$(iii) \tan(360^\circ - \theta) = -\tan\theta$$

$$(v) \sec(360^\circ - \theta) = \sec\theta$$

$$(ii) \cos(360^\circ - \theta) = \cos\theta$$

$$(iv) \cot(360^\circ - \theta) = -\cot\theta$$

$$(vi) \operatorname{cosec}(360^\circ - \theta) = -\operatorname{cosec}\theta$$

(m) T-functions of angle “ $360^\circ + \theta$ ” in terms of “ θ ”

$$(i) \sin(360^\circ + \theta) = \sin\theta$$

$$(iii) \tan(360^\circ + \theta) = \tan\theta$$

$$(v) \sec(360^\circ + \theta) = \sec\theta$$

$$(ii) \cos(360^\circ + \theta) = \cos\theta$$

$$(iv) \cot(360^\circ + \theta) = \cot\theta$$

$$(vi) \operatorname{cosec}(360^\circ + \theta) = \operatorname{cosec}\theta$$

(n) T-functions of angle “ $n \times 360^\circ + \theta$ ” in terms of “ θ ”

$$(i) \sin(n \times 360^\circ + \theta) = \sin\theta$$

$$(iii) \tan(n \times 360^\circ + \theta) = \tan\theta$$

$$(v) \sec(n \times 360^\circ + \theta) = \sec\theta$$

$$(ii) \cos(n \times 360^\circ + \theta) = \cos\theta$$

$$(iv) \cot(n \times 360^\circ + \theta) = \cot\theta$$

$$(vi) \operatorname{cosec}(n \times 360^\circ + \theta) = \operatorname{cosec}\theta$$

Where “n” is the angle of revolution.

4. T-functions of compound angle

$$(i) \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$(iii) \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$(v) \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$(vii) \cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$$

$$(ii) \sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$(iv) \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$(vi) \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$(viii) \cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

$$(ix) \tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

5. Transformation formula

$$(a) (i) \quad \sin C + \sin D = 2 \sin \frac{C+D}{2} \cdot \cos \frac{C-D}{2}$$

$$(ii) \quad \sin C - \sin D = 2 \cos \frac{C+D}{2} \cdot \sin \frac{C-D}{2}$$

$$(iii) \quad \cos C + \cos D = 2 \cos \frac{C+D}{2} \cdot \cos \frac{C-D}{2}$$

$$(iv) \quad \cos C - \cos D = 2 \sin \frac{C+D}{2} \cdot \sin \frac{D-C}{2}$$

$$(b) (i) \quad 2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$(ii) \quad 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$(iii) \quad 2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$(iv) \quad 2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

6. T-functions of multiple angle

(a)

$$(i) \quad \sin 2A = 2 \sin A \cos A$$

$$(ii) \quad \cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\boxed{2 \sin^2 A = 1 - \cos 2A}$$

$$\boxed{2 \cos^2 A = 1 + \cos 2A}$$

$$(iii) \quad \tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$(b)(i) \quad \sin 2A = \frac{2 \tan A}{1 + \tan^2 A} \quad (ii) \quad \cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$(c) (i) \quad \sin 3A = 3 \sin A - 4 \sin^3 A \quad (ii) \quad \cos 3A = 4 \cos^3 A - 3 \cos A$$

$$(iii) \quad \tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

7. T-functions of submultiple angle

(a)

$$(i) \quad \sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$(ii) \quad \cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = 2 \cos^2 \frac{A}{2} - 1 = 1 - 2 \sin^2 \frac{A}{2}$$

$$\boxed{2 \sin^2 \frac{A}{2} = 1 - \cos A}$$

$$2c \cos^2 \frac{A}{2} = 1 + \cos A$$

$$(iii) \tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}$$

$$(b)(i) \sin A = \frac{2 \tan \frac{A}{2}}{1 + \tan^2 \frac{A}{2}} \quad (ii) \cos A = \frac{1 - \tan^2 \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}$$

8. Trigonometrical equations and its general solutions.

$$1. \sin \theta = 0 \Rightarrow \theta = n\pi$$

$$2. \cos \theta = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$$

$$3. \tan \theta = 0 \Rightarrow \theta = n\pi$$

$$4. \sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$$

$$5. \tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$$

$$6. \cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$$

Where n is an integer or, $n \in I$

9. Properties of triangle

1. Sine rule

$$(i) \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$(ii) \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

2. Tangent rule

$$(i) \tan \frac{A-B}{2} = \frac{a-b}{a+b} \cot \frac{C}{2}$$

$$(ii) \tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$(ii) \tan \frac{C-A}{2} = \frac{c-a}{c+a} \cot \frac{B}{2}$$

3. Cosine Formula

$$(i) \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$(ii) \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$(iii) \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

4. projection formula (i) $a = b \cos C + c \cos B$ (ii) $b = a \cos C + c \cos A$

$$(iii) c = a \cos B + b \cos A$$

5. Some other formulae

$$(i) \sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$$

$$(ii) \sin \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$$

$$(iii) \sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

$$(iv) \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$$

$$(v) \quad \cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$$

$$(vii) \quad \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

$$(ix) \quad \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$(xi) \quad \cot \frac{B}{2} = \sqrt{\frac{s(s-b)}{(s-a)(s-c)}}$$

$$(vi) \quad \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

$$(viii) \quad \tan \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$$

$$(x) \quad \cot \frac{A}{2} = \sqrt{\frac{s(s-a)}{(s-b)(s-c)}}$$

$$(xii) \quad \cot \frac{C}{2} = \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

6. Area Formulae

$$(i) \quad \Delta = \frac{1}{2}ab \sin C = \frac{1}{2}ac \sin B = \frac{1}{2}bc \sin A$$

$$(ii) \quad \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$(iii) \quad \tan \frac{A}{2} = \frac{\Delta}{s(c-a)}, \tan \frac{B}{2} = \frac{\Delta}{s(s-b)}, \tan \frac{C}{2} = \frac{\Delta}{s(s-c)}$$

$$\text{Where, } \Delta = \text{Area of triangle ABC and } s = \frac{a+b+c}{2}$$

10. Inverse circular functions:

$$\text{a. (i) } \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad \text{(ii) } \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \quad \text{(iii) } \operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$$

$$\text{b. (i) } \sin^{-1} x = \operatorname{cosec}^{-1} \frac{1}{x} \quad \text{(ii) } \cos^{-1} x = \sec^{-1} \frac{1}{x} \quad \text{(iii) } \tan^{-1} x = \cot^{-1} \frac{1}{x}$$

$$\text{c. (i) } \sin(\sin^{-1} x) = x \quad \text{(ii) } \cos(\cos^{-1} x) = x \quad \text{(iii) } \tan(\tan^{-1} x) = x$$

$$\text{(iv) } \cot(\cot^{-1} x) = x \quad \text{(iv) } \sec(\sec^{-1} x) = x \quad \text{(v) } \operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$$

$$\text{d. (i) } \sin^{-1}(\sin x) = x \quad \text{(ii) } \cos^{-1}(\cos x) = x \quad \text{(ii) } \tan^{-1}(\tan x) = x$$

$$\text{(iv) } \cot^{-1}(\cot x) = x \quad \text{(v) } \sec^{-1}(\sec x) = x \quad \text{(vi) } \operatorname{cosec}^{-1}(\operatorname{cosec} x) = x$$

$$\text{e. (i) } \sin^{-1}(-x) = -\sin^{-1} x \quad \text{(ii) } \cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$\text{(iii) } \tan^{-1}(-x) = -\tan^{-1} x \quad \text{(iv) } \cot^{-1}(-x) = \pi - \cot^{-1} x$$

$$\text{f. (i) } \sin^{-1} x + \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$$

$$\text{(ii) } \sin^{-1} x - \sin^{-1} y = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$$

$$\text{(iii) } \cos^{-1} x + \cos^{-1} y = \cos^{-1}(xy - \sqrt{(1-x^2)(1-y^2)})$$

$$\text{(iv) } \cos^{-1} x - \cos^{-1} y = \cos^{-1}(xy + \sqrt{(1-x^2)(1-y^2)})$$

$$\text{(v) } \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \quad \text{(vi) } \tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$$

$$\text{(vi) } \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left(\frac{x+y+z-xyz}{1-xy-yz-zx} \right)$$

g. (i) $2 \sin^{-1} x = \sin^{-1}(2x\sqrt{1-x^2})$

(ii) $2 \cos^{-1} x = \cos^{-1}(x^2 - \sqrt{1-x^2}) = \cos^{-1}(2x^2 - 1)$

h.(i) $2 \tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

i. (i) $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$ (ii) $3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x)$

(iii) $3 \tan^{-1} x = \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right)$

IMPORTANT RESULTS

(i) $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

(ii) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

THE END